An Explainable AI Model for IoMT Network Attack Detection

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Abstract

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1 Introduction

Smart and connected medical devices are increasingly integrated into the delivery of healthcare services, leading to the development of the Internet of Medical Things (IoMT), a subset of the Internet of Things (IoT) encompassing medical equipment and devices. IoMT technology offers many advantages by enabling continuous monitoring of patients' health. This assists healthcare professionals in making remote diagnoses and better informing their clinical decisions; for example, data collected from IoMT devices has been used to predict patients' risk of heart disease [1]. With the growing digitalization of healthcare systems, there has been a significant increase in cyberattacks targeting medical infrastructure. According to the HIPAA Journals' report, there have been 546 healthcare breaches that affect more than 500 people so far in 2025 [2]. This, coupled with the sensitive nature of medical data, raises major concerns about patients' privacy, where patients' personal data may be exposed, as well as potential concerns about patients' safety, given that IoMT devices are used to deliver treatments. Recent research has shown that Machine Learning (ML) can be effectively used to detect network attacks by analyzing network traffic. Compared to traditional network attack detection methods, such as rule-based systems, ML techniques offer significant improvements by recognising complex patterns in network traffic that were previously not possible. However, the problem with complex ML techniques is their lack of interpretability in decision-making; in the context of healthcare systems, this interpretability is crucial, as medical staff and patients need to trust that IoMT networks are secure. To address the lack of interpretable ML models, this project will develop an Explainable Artificial Intelligence (XAI) model to detect network attacks in IoMT systems. This ensures that users of the system can understand and trust the model while still achieving high detection accuracy. This project seeks to design and implement an XAI model that can detect network attacks in an IoMT system, whilst maintaining interpretability in its detections. The proposed model will be trained and evaluated using the CIC2024 IoMT dataset, a publicly available dataset synthesised by the Canadian Institute for Cybersecurity [3]. This dataset has been specifically designed to develop and benchmark network attack detection models. It simulates a number of IoMT devices and a wide variety of both benign and malicious network traffic.

2 Literature Review

2.1 IoMT Architecture and Security Concerns

IoMT systems are typically divided into three primary layers: perception, network, and applications. [4]. In the perception layer, we have the actual medical sensors and wearable devices that collect information from patients, such as tracking cardiovascular and heart health, temperature, and sleep patterns. This layer faces potential vulnerabilities, with the potential for tampering and false data collection, which could ultimately lead to incorrect diagnoses based on incorrect data. The network layer enables these devices to communicate with other devices in the IoMT network using protocols such as Wi-Fi and Bluetooth. This layer is susceptible to attacks such as DoS and DDoS, which can lead to disruptions in treatment and pose an immediate risk to patients' safety when IoMT devices are unable to operate. Man-in-the-middle attacks at this layer can also lead to data breaches, compromising the confidentiality of sensitive patient and medical staff data. Moving to the application layer, we find software vulnerabilities where cyber threats, such as phishing and ransomware, can lead to interruptions in medical care and further data breaches. Across these layers, there are three clear points of risk for IoMT technology: patients' data privacy, patients' safety, and disruption to medical systems. The variety of areas that are vulnerable in these systems highlights the need for a more rigorous and trustworthy cybersecurity system to reduce the number of successful attacks. Effective Intrusion Detection System (IDS) is an essential tool for preventing the wide variety of attacks that can IoMT networks face.

2.2 ML-Driven IDS in IoMT Environments

Traditional Intrusion Detection Systems (IDS) such as signature based methods are not well-suited to the field of IoT networks, this has lead to research in recent years primarily being focused on hybrid Machine Learning (ML) methods for detecting network attacks, including Random Forest and SVM [5]. These ML-based systems aim to analyse large amounts of network traffic and distinguish between benign and malicious activity. In this section, I will focus on reviewing the most well-documented models used and how their performance is evaluated in IoT and IoMT systems. Continuing with the shift toward ML-driven IDS approaches, early IDS explored the efficacy of algorithms such as Decision Trees and SVMs, both of which achieve promising results when tested on benchmark datasets. For example, a Hybrid Decision Tree tested and evaluated on the NSL-KDD dataset [6] achieves 83% accuracy and 0.83 F1 score, outperforming other ML methods such as SVM and KNN [7]. However, this model does not distinguish between attack types such as DoS, MITM, and others, which is desirable for IoMT security; the authors suggest future work addresses this limitation. Deep learning methods provide a solution to this, a deep reinforcement learning model was developed and testd alongside traditional ML methods to detect malicious network traffic in IoMT networks. Using the NSL-KDD dataset [6], the best performing model was the Double Deep Q-Network (DDQN) achieving an accuracy of around 80% and F1 score of 0.91[8]. Comparing this the Hybrid Decision Tree, it is clear that this form of deep reinforcement learning excels in attack detection. A noticeable drawback of deep learning here is the time taken to train the model compared to traditional ML as well as the lack of interpretability it provides.

2.3 Explainable Artificial Intelligence (XAI) in IDS

Explainable Artificial Intelligence (XAI) refers to a set of tools which aim to make the predictions made by a deep learning model understandable to humans, this is in the form of showing the features the model decides played the biggest role in making the final prediction. Obviously, in a healthcare setting, the ability to trust is essential with the IDS system securing an IoMT network because we do not want to risk genuine medical communications being interrupted, which could lead to disruptions to patients treatment. However, we still want to accurately prevent malicious attacks that could cause reduced patient safety and sensitive medical data being compromised. Therefore, integrating XAI into a deep learning IDS allows the healthcare industry to understand and validate a the predictions of the model. The two main types of XAI models are model-specific and model-agnostic. Model specific XAI is suited to a specific group of models, meaning they offer a deep insight into the models decision making, however they cannot be applied to a wide variety of model types. Model agnostic techniques are the opposite, they can be applied to many model types with a less in-depth insight. [9]. Some examples of XAI techniques are LIME and SHAP. LIME (Local Interpretable Mode-agnostic Explanations) is an algorithm proposed in 2016, it explains the prediction of a classifier (model-agnostic) though approximating it with a simpler and interpretable model [10]. Later, SHAP (Shapley Additive exPlanations) was introduced in 2017 [11], similar to LIME, it is a model-agnostic algorithm that determines how each feature contributes to a models output. Recent studies and research demonstrate that XAI models, such as LIME and SHAP are effective in explaining predictions made by IDS systems. In [12], researchers successfully introduce an explainable machine learning framework for IDS, this framework was built around the SHAP method and used the NSL-KDD dataset [6] to evaluate the models performance. However, this paper mentions the frameworks limitations and highlights the need for more datasets to be tested to improve the explainability. A deep learning based IDS is proposed in [13], the authors integrate XAI models including both LIME and SHAP. This paper evaluates the performance on both the NSL-KDD dataset as well as UNSW-NB15, demonstrating that XAI models make these deep learning models more transparent and trustworthy. There is limited work in applying XAI techniques specifically to IoMT scenarios, the trade-off experienced between interpretabiltity and latency remains under-researched. Aiming to address this gap will allow the development of more trustworthy IDS models in healthcare settings that use IoMT networks.

2.4 Evalution Datasets and Benchmarks

The accuracy and reliability of Machine Learning (ML) based Intrusion Detection Systems (IDS) are primarily dependent on the quality and the realism of the data used to train, validate and test the model. This is especially important in IoMT settings because the data transmitted in these networks are fundamentally more sensitive than a typical IoT network, making the need for secure communications essential. This section will critically evaluate the available datasets that have previously been used in ML-driven IDS models, discuss the limitations and justify the selection of the CIC2024 IoMT Dataset for this projects specific focus. The dataset, NSL-KDD [6], frequently mentioned in the previous sections of this literature review, has been used extensively in training IDS models. This dataset was synthesised to improve upon its predecessor and has been able to produce promising results in many research application. However, this dataset has drawbacks such as its high redundancy of its records, which can lead to poor generalization in ML models [14]. This dataset would not be suitable for the model proposed in this report because it does not reflect an accurate IoMT system. Newer datasets provide significant improvements of NSL-KDD, one notable example is the CICIDS2017 dataset, this dataset includes more realistic network traffic in IoT networks. However, this would also not be suited to IoMT systems due to it not containing common examples of network traffic such as lightweight protocols MQTT and Bluetooth which are used by IoMT devices. To maximise the realism of the data used to train the proposed model, this project will use the CIC2024 IoMT Dataset [3]. This dataset was specifically synthesised for benchmarking security systems for IoMT. It involves traffic types missing from the previously mentioned datasets including common protocols used by IoMT such as MQTT and Bluetooth. This dataset also provides a wide variety of current and relevant attack vectors faced by real world IoMT networks. The structure of the dataset will guide the

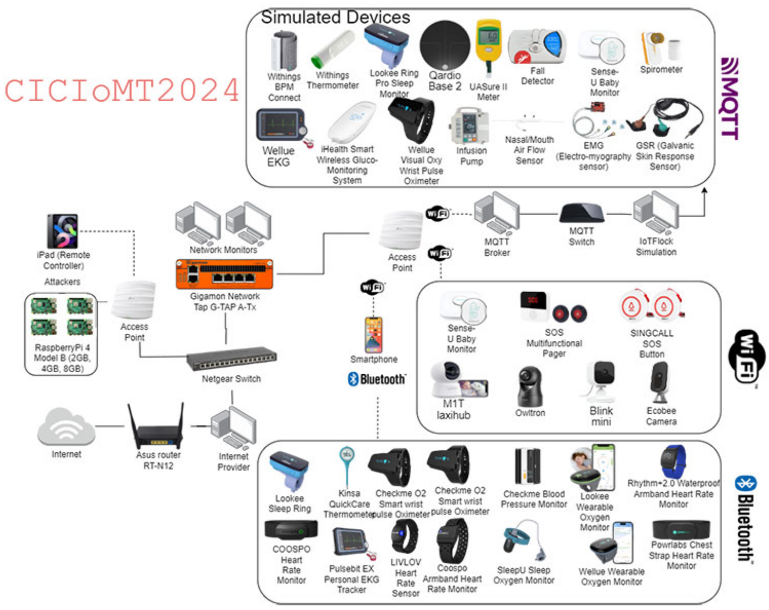


Figure 1: CICIoMT2024 topology. Source: [3]

development of this model. Due to the large class imbalance between benign and malicious network traffic, the evaluation of the model will need to incorporate various metrics such as F1-score, Recall, and Precision.

3 Project Specification

3.1 Motivation

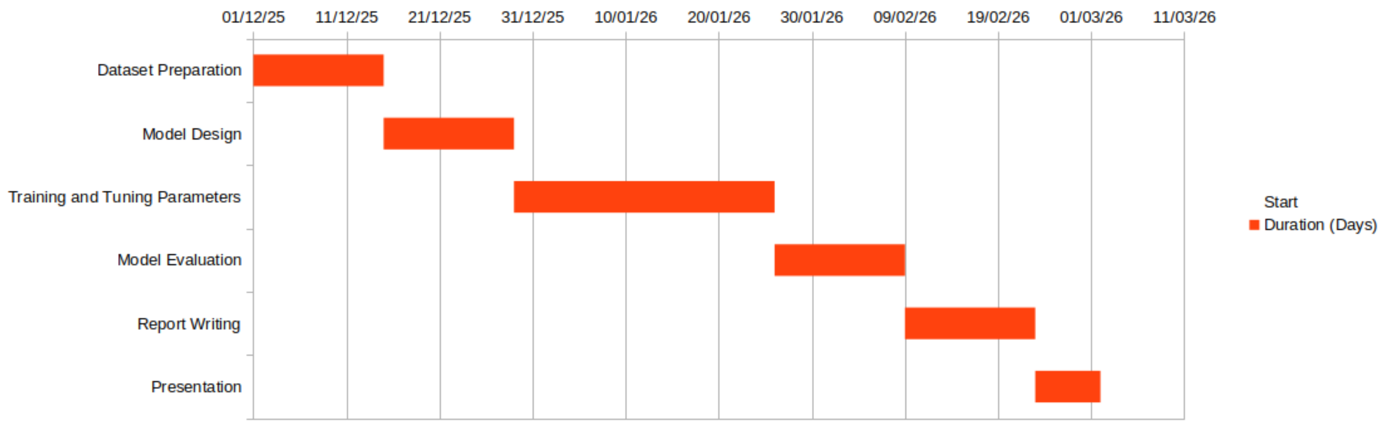
The use of IoMT systems continues to spread in the field of modern healthcare, with many routin health checks now able to be performed remotely and continuously. Unfortunately, with this increase of its use, there has also been an increase in the amount of cyber-attacks focused on healthcare services [2]. The main goal and motivation of this project is to address the lack of trust that can be placed in Machine Learning based Intrusion Detection Systems, through developing an explainable model who's decisions can be verified and understood.

3.2 Objectives

The primary objectives I will follow this project are outlined as followed:

- Review current literature on AI-based network attack detection within IoT and IoMT networks
- Develop an accurate and interpretable attack detection model and select a suitable XAI technique
- Evaluate the models performance using the CIC2024 IoMT dataset
- Assess the models interpretability and its suitability for cybersecurity specifically to a healthcare context

3.3 Gantt Chart Timeline



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